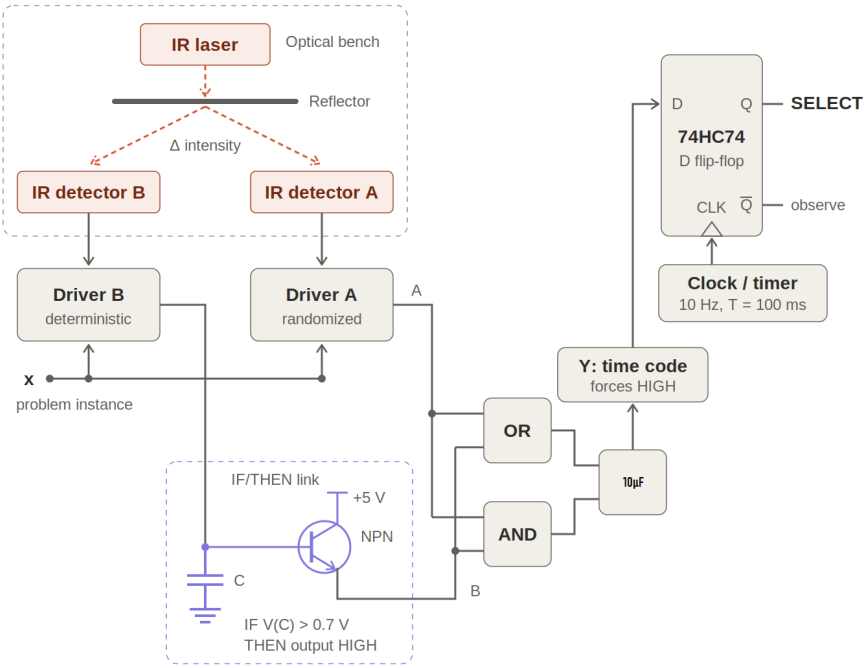
PART 1 OF 3

How the RP-Style Decision Circuit Works

A walkthrough of the optoelectronic version: optical bench, parallel drivers A and B, IF/THEN link, OR/AND gating, time code Y and a 74HC74 flip-flop



The circuit as supplied

1. The big idea

The circuit is a hardware picture of a **Randomized Polynomial-time (RP)** decision procedure. RP is the class of yes/no problems for which a probabilistic algorithm has **one-sided error**: if the answer is NO (x is not in L) the algorithm never says yes, and if the answer is YES (x is in L) it says yes with probability at least 1/2. The circuit runs one such trial. An optical bench (an IR laser, a reflector and two IR detectors) supplies the light signals. The problem instance x and those signals go to two drivers in parallel, the driver outputs are combined by logic gates, and the verdict is stored in a flip-flop on a clock edge. That stored bit, **SELECT = Q**, is the answer.

Reading note. This drawing labels the drivers only as “randomized” and “deterministic”. The RP properties used below (A is always 0 when x is not in L and is 1 with probability at least 1/2 when x is in L; B is 1 whenever x is in L) are the standard reading carried over from the earlier version of the figure. They are assumptions about what the drivers do, not text printed on this drawing.

2. The blocks and what each one does

Block	Role	Behaviour
IR laser	Light source	Emits the IR beam. Above threshold a laser gives close to Poissonian photon counts (Fano factor 1), with the mean set by the drive current.
Reflector	Beam steering / split	Sends the beam to both detectors. The label “Δ intensity” is read as the difference between the light reaching detector A and detector B. The drawing does not say how it is set.

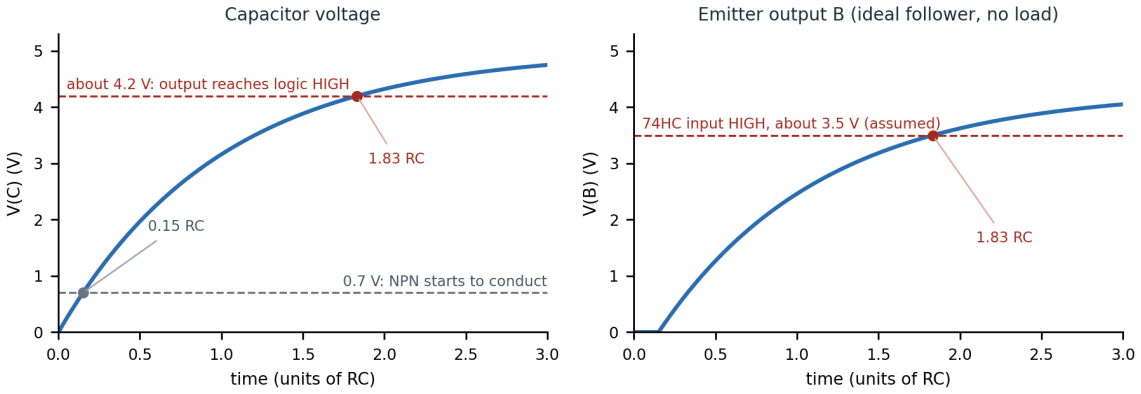
Block	Role	Behaviour
IR detectors A and B	Light to electrical signal	Each turns the light it receives into a photocurrent. Detector A feeds Driver A and detector B feeds Driver B. They see the same beam, so laser intensity noise is common to both while shot noise is independent.
Driver A (randomized)	Probabilistic witness test	Uses its noisy detector signal as the random source to try a random witness r and test w(x, r). If x is in L, A = 1 with probability ≥ 1/2. If x is not in L, A = 0 always (no witness can exist).
Driver B (deterministic)	Cheap precondition	A fixed check c(x) with no randomness. B = 1 whenever x is in L. Its output does not go straight to the gates: it charges the capacitor in the IF/THEN link.
IF/THEN link (C and NPN)	Charge threshold	Driver B's output charges capacitor C. When V(C) exceeds about 0.7 V (one silicon base-emitter drop) the NPN conducts from the +5 V rail and its emitter output, labelled B, goes HIGH. In words: IF the capacitor holds enough charge, THEN output B.
OR gate	Either driver can contribute	Outputs 1 when A or B (or both) is 1.
AND gate	Agreement check	Outputs 1 only when A = 1 and B = 1, i.e. the random witness test succeeded and the deterministic check agrees.
10 µF node	Smoothing / hold	The two gate outputs meet at the block labelled 10 µF, read as the smoothing capacitor of the earlier drawing (the capacitor symbol and its ground connection are no longer drawn). It low-pass filters glitches so the signal is stable when the clock samples it.
Time code Y	Timing gate	Labelled "forces HIGH". It sits between the 10 µF node and the D input. The bit pattern is not printed on this drawing (the earlier one showed 1011). Section 7 explains how "forces HIGH" is read.
Clock / timer	Sampling instant	10 Hz (T = 100 ms). It carries no data, it just tells the flip-flop when to look at D.
74HC74 D flip-flop	Memory of the verdict	On the rising CLK edge it copies D to Q and holds it. Q is SELECT (the decision). The inverted output, Q-bar, is labelled "observe".

3. Step-by-step signal flow

1. **Light.** The IR laser illuminates the reflector, which sends light to detectors A and B. The intensity difference between the two is the optical signal.
2. **Detect.** Each detector converts its light into an electrical signal for its own driver.
3. **Input.** The instance x is fanned out to both drivers at once (the black dot at the left).
4. **Drivers fire in parallel.** A uses its noisy signal to try a witness. B runs its fixed check. Each produces one result.
5. **IF/THEN.** Driver B's output charges C. Once V(C) passes about 0.7 V the NPN turns on and B goes HIGH.
6. **Gates combine them.** A (straight from Driver A) and B (from the NPN) each go to both the OR and the AND gate. The earlier drawing's caption required two intermediate conditions, (A OR B) and (A AND B). This drawing has no caption and that reading is kept.
7. **Node and time code.** The gate outputs join at the 10 µF node, which smooths them, and the time code Y presents a defined HIGH to the flip-flop's D pin.
8. **Clock edge.** Every 100 ms the rising edge on CLK makes the 74HC74 latch whatever is on D.
9. **Result.** Q = SELECT = 1 means "accept: x is in L". Q = 0 means "no witness found". Q-bar ("observe") shows the opposite polarity.

4. The IF/THEN link in detail

Driver B's output charges capacitor C, and the NPN watches the voltage on C. Suppose Driver B drives a voltage V_s through a resistance R. Then $V(C) = V_s\,(1 - e^{-t/RC})$, and the transistor starts to conduct when V(C) reaches about 0.7 V, at $t = RC \cdot \ln[V_s / (V_s - 0.7\ V)]$. For V_s = 5 V this is 0.15 RC.



Left: capacitor voltage for a 5 V drive through resistance R. Right: emitter output B of an ideal follower with no load, against an assumed 3.5 V logic-HIGH level.

The NPN is wired as an emitter follower: collector on +5 V, emitter as the output. Its emitter voltage therefore follows V(C) minus about 0.7 V rather than snapping to the rail. A 74HC input at 5 V needs roughly 3.5 V to read as HIGH, so B only becomes a valid logic HIGH once V(C) is near 4.2 V, which takes 1.83 RC in this example. The box's rule "IF V(C) > 0.7 V THEN output HIGH" is the turn-on condition of the transistor. It is not the moment B becomes a clean logic level.

Two consequences follow. **Timing:** B can only be 1 in time if the clock edge arrives after the capacitor has charged, so RC has to sit well inside the 100 ms clock period. **Statistics:** the capacitor adds up many electrons before the threshold is crossed, which averages out fast fluctuations. That averaging is what lets Driver B behave deterministically, as Part 2 explains.

5. Why the error is one-sided

Q = 1 only if both A OR B and A AND B hold, which collapses to A AND B (since A AND B already implies A OR B). So the circuit accepts only when the randomized driver finds a witness and the deterministic branch agrees in time:

Case	A	B	SELECT (Q)	Meaning
x not in L	always 0	any	0	Never a false accept. No error possible.
x in L, witness found, link charged in time	1	1	1	Correct accept (probability ≥ 1/2).
x in L, witness missed	0	1	0	False reject (probability ≤ 1/2).
x in L, witness found, link too slow	1	0	0	Extra false reject from a timing failure in the IF/THEN link. Safe, but it lifts the error above 1/2 unless the link settles before the clock edge.

An instance outside L never gives SELECT = 1. A wrong answer can only be a missed YES, never a fake YES. The IF/THEN link can add missed YES cases but cannot create a false accept, because the AND needs A = 1 and A is 0 outside L. That last point holds for an ideal Driver A; a noisy physical detector weakens it (see Part 2, section 5).

6. Repeating trials to push the error down

One trial can wrongly reject a YES instance with probability up to 1/2. Because trials are independent and a single success is conclusive, you run the circuit k times and accept if any trial gives SELECT = 1. The chance of every trial failing is at most $(1/2)^k$. At the 10 Hz clock, k trials take k × 100 ms.

Trials k	Worst-case error	Time at 10 Hz
1	0.5	0.1 s
2	0.25	0.2 s
5	0.03125	0.5 s
10	0.000976562	1.0 s
20	9.54e-07	2.0 s

Two practical points. The 74HC74 overwrites Q at every clock edge, so “accept if any trial succeeds” needs an OR over the successive Q values (or a set-only latch). And C has to be discharged before each trial, otherwise it can still hold charge from the last one.

7. Things to be aware of in the drawing

The figure is a conceptual illustration, not a build-ready schematic. If you ever wire it up, check these points:

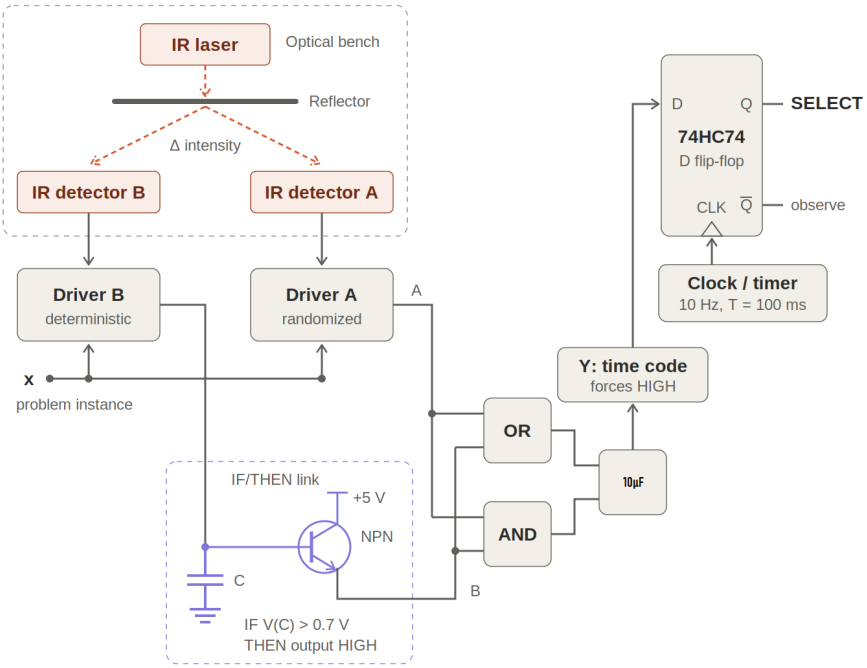
- **How the two gates combine.** AND implies OR, so requiring both (A OR B) and (A AND B) reduces to A AND B. Tied together as drawn, the two outputs would instead behave like a plain OR (A OR B). The “both conditions” reading is the one that keeps the logic meaningful.
- **Tied gate outputs.** Directly joining two CMOS outputs causes bus contention when they disagree. A real design would use an AND gate (or a proper combining stage) to merge them.
- **A OR B alone would break the guarantee.** If B can be 1 for an x outside L, accepting on A OR B could give a false accept. Only the AND path preserves one-sided error.
- **The IF/THEN link is a threshold, not a logic level.** See section 4. The emitter also has no pull-down, so when the NPN is off the output floats instead of reading LOW. There is no base resistor and no path to discharge C between trials.
- **“Forces HIGH”.** Read literally, a time code that forces D HIGH unconditionally would make Q equal 1 at every clock edge whatever A and B did, and the circuit would carry no information. The reading used here is that Y marks the time bins in which the gate result is allowed through to D. The drawing does not settle this.
- **Shared beam.** Both detectors see the same laser. Laser intensity noise is common to A and B, so the drivers’ inputs are not fully independent. That matters if the RP argument needs A’s randomness to be independent of B.
- **Noise on the A side.** A real detector signal has noise and dark current, so A could fire when no witness exists. “A = 0 always outside L” is then only approximate.
- **74HC74 housekeeping.** Real use needs the PRE and CLR pins tied high, VCC and GND connected, and decoupling near the chip. These are not shown.
- **Q-bar is not extra information.** It is simply NOT Q, labelled “observe”, useful for an indicator LED or active-low logic.

Summary: light reaches two detectors, two drivers test x in parallel, B passes through a charge-threshold link, the gates demand agreement, a clocked D flip-flop latches the verdict, and repetition drives the false-reject rate toward zero while false accepts stay impossible in the ideal case.

PART 2 OF 3

From Electrons to Photon Statistics

Re-reading the optoelectronic RP-style circuit as a model of how electron handling shapes the light a device emits and detects



The source circuit: optical bench, drivers A and B, IF/THEN link, OR/AND gating, time code Y and a 74HC74 flip-flop

Description of the circuit. An IR laser sends light via a reflector to two IR detectors. Detector A feeds Driver A, a randomized witness test: if x is in L it outputs 1 with probability at least 1/2, and if x is not in L it always outputs 0. Detector B feeds Driver B, a deterministic check that outputs 1 whenever x is in L; its output charges a capacitor C, and when V(C) exceeds about 0.7 V an NPN transistor passes B HIGH. A and B go to an OR gate and an AND gate, meet at a 10 µF node, and pass through the time code Y ("forces HIGH") to the D input of a 74HC74 flip-flop. A 10 Hz clock tells the flip-flop when to sample, and Q (SELECT) holds the one-bit verdict, with Q-bar as the observe output. Because A is always 0 outside L, SELECT can never be 1 for a NO instance: the error is one-sided, and repeating trials shrinks the remaining error as $(1/2)^k$. (The RP properties of A and B are the assumed reading described in Part 1.)

1. The claim

In electronics and optoelectronics, **the way electrons are manipulated determines the statistical distribution of the photons emitted**, and the statistics of the photons that arrive determine the statistics of the electrons detected. If electrons arrive at a junction as a random, uncorrelated stream, the photons come out random too. If the electrons are metered out one at a time, the photons inherit that regularity. This circuit has both ends of that chain: the laser turns electrons into photons, and the detectors turn photons back into electrons. The RP circuit is a convenient frame for this: it has a random branch, a deterministic branch, a charge threshold, a gating stage, a smoothing capacitor, a timing code and a clocked latch. Each has a physical counterpart in a light-emitting or light-detecting device.

2. Three numbers that describe photon statistics

Quantity	Definition	Reading
Fano factor F	$F = \mathrm{Var}(n) / \langle n \rangle$ for photons counted in a window	$F = 1$ Poissonian (laser, random stream). $F > 1$ super-Poissonian (thermal, bunched). $F < 1$ sub-Poissonian (regulated, antibunched).
Mandel Q	$Q = F - 1$	$Q = 0, > 0, < 0$ for the three cases above. Only $Q < 0$ has no classical-wave explanation.
$g^{(2)}(0)$	Probability of two photons arriving together, relative to chance	Thermal light: 2. Laser: 1. Ideal single-photon emitter: 0.

3. Re-labelling the circuit

Circuit element	Electronic / optoelectronic counterpart	Statistical effect
Input x	The setting applied to both drivers (bias, threshold, drive level) that fixes the mean signal	Sets the mean photon or electron number.
IR laser	Electrons injected into a diode laser recombine and emit photons. Above threshold the output is coherent.	Poissonian with a random pump. Sub-Poissonian if the pump current is regulated (amplitude-squeezed light).
Reflector (Δ intensity)	A passive splitter that sends part of the beam to each detector. The intensity difference between the two arms is the signal.	Splitting is random thinning, which pushes each arm toward Poisson. Sub-Poissonian light gives anticorrelated counts in the two arms.
IR detectors A and B	Photodiodes that turn absorbed photons into electrons with efficiency η_d	Thin the stream again. Electron counts inherit the photon statistics, scaled by η_d.
Driver A (random oracle / noise source)	Thresholding a noisy photocurrent: shot noise, thermal noise and dark current act as the random oracle	Random outcome, near Poissonian or, with thermal mixing, super-Poissonian.
Driver B (deterministic check)	Regulated drive, with the signal averaged on the capacitor in the IF/THEN link. On the emitting side the analogue is a constant-current drive through a high impedance, a single-electron turnstile or Coulomb blockade in a quantum dot.	Regular arrival, so the outcome is nearly deterministic and, on the emitting side, sub-Poissonian.
IF/THEN link (C and NPN)	A charge-integrating threshold, like an integrate-and-fire stage. It needs $N = C \times 0.7\ \mathrm{V} / e$ electrons before the NPN conducts.	Averaging over N electrons cuts relative fluctuations to about $1/\sqrt{N}$. For $C = 1$ pF, $N \approx 4.4 \times 10^6$ and $1/\sqrt{N} \approx 0.05\%$.
AND gate	Coincidence condition: both channels must signal together (in an emitter, an electron and a hole must both be present before radiative recombination)	Output only if both conditions hold; gates the result.
OR gate	Either the random or the regulated channel can trigger the output	Widens acceptance. On its own it would let noise through.
10 µF node	Circuit capacitance with series resistance; filters fast fluctuations (in a single-electron device, the charging energy)	Sets the bandwidth over which noise is smoothed or blocked.
Time code Y	A gating pattern (pump pulses or detector gates) marking the time bins in which the result may pass	Fixes when counts may appear, shaping the temporal statistics.

Circuit element	Electronic / optoelectronic counterpart	Statistical effect
Clock, 10 Hz	Pulsed excitation or detector gating (real devices: kHz to GHz)	Defines the counting window; 10 Hz is symbolic.
74HC74 D flip-flop, Q = SELECT	Threshold discriminator and latch: a “click”	Converts the stream into a recorded 0/1 outcome per window.
Q-bar (observe)	The complementary “no click” record	Same information, opposite polarity.

4. How electron handling sets the photon distribution

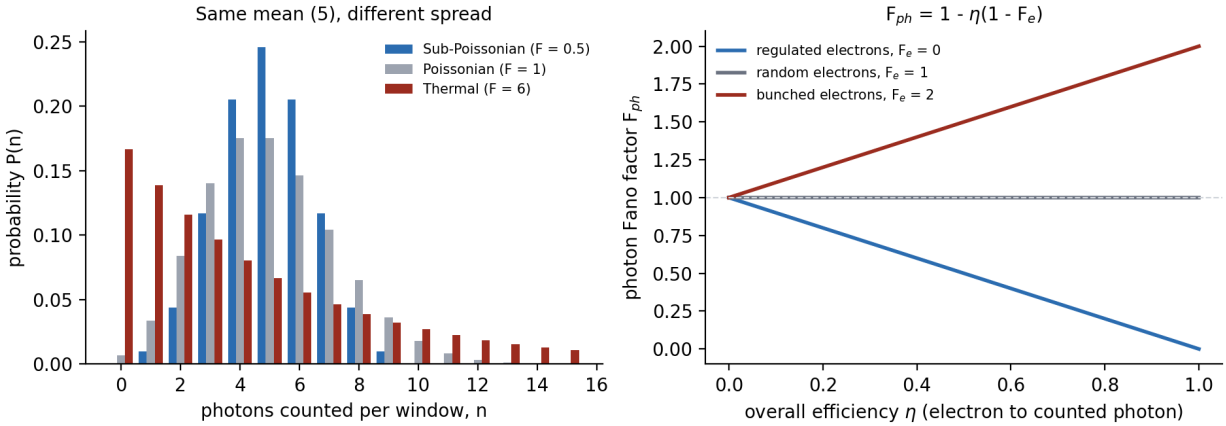
Suppose each injected electron produces a counted photon with overall probability η (internal quantum efficiency times extraction times detector efficiency), independently of the others. Random thinning of a stream gives a simple relation between the electron Fano factor F_e and the photon Fano factor F_{ph}:

$$F_{ph} = 1 - \eta\,(1 - F_e)$$

Three regimes follow directly:

- **Random electrons ($F_e = 1$)**, the Driver A case: $F_{ph} = 1$ whatever η is. The photons are Poissonian.
- **Regulated electrons ($F_e = 0$)**, the Driver B case: $F_{ph} = 1 - \eta$. With η = 0.8 the photon stream has F = 0.2, strongly sub-Poissonian. The regularity survives only as far as the efficiency allows.
- **Bunched electrons ($F_e > 1$)**, for example trap-limited or avalanche-like injection: $F_{ph} > 1$, super-Poissonian.

Detection runs the same relation backwards. A detector with efficiency η_d thins the photon stream in the same way, so the electron counts it produces have $F_{det} = 1 - \eta_d\,(1 - F_{ph})$. Losses in series multiply, so from pump electrons to detector electrons $F_{det} = 1 - \eta_{emit}\,T\,\eta_d\,(1 - F_e)$, where T is the share of the light the reflector sends to that detector. For example, a regulated pump ($F_e = 0$) with $\eta_{emit} = 0.8$, T = 0.9 and $\eta_d = 0.7$ gives an overall η of 0.504 and $F_{det} \approx 0.50$. Half of the pump’s regularity is gone by the time the signal reaches Driver B.



Left: photon-number distributions with the same mean. Right: the Fano-factor relation above.

5. One-sided error becomes detector behaviour

The circuit’s guarantee was: *no witness, no accept*. The detector on branch A behaves the same way. With no signal there is (apart from dark counts and noise) no click, so false accepts are essentially absent. A signal that is present can still be missed, with probability 1 − η, which plays the role of the false-reject chance (≤ 1/2 in the RP definition, so it needs η ≥ 1/2 to match).

Repeating trials has a direct counterpart too: counting over k clock windows drives the chance of missing a real signal down to $(1 - \eta)^k$. With η = 0.5, ten windows leave about 0.1%. Noise and dark counts are the one place the analogy breaks cleanly. They can fire Driver A when no witness exists, which makes them false accepts, something the ideal RP circuit never has, and they accumulate as you

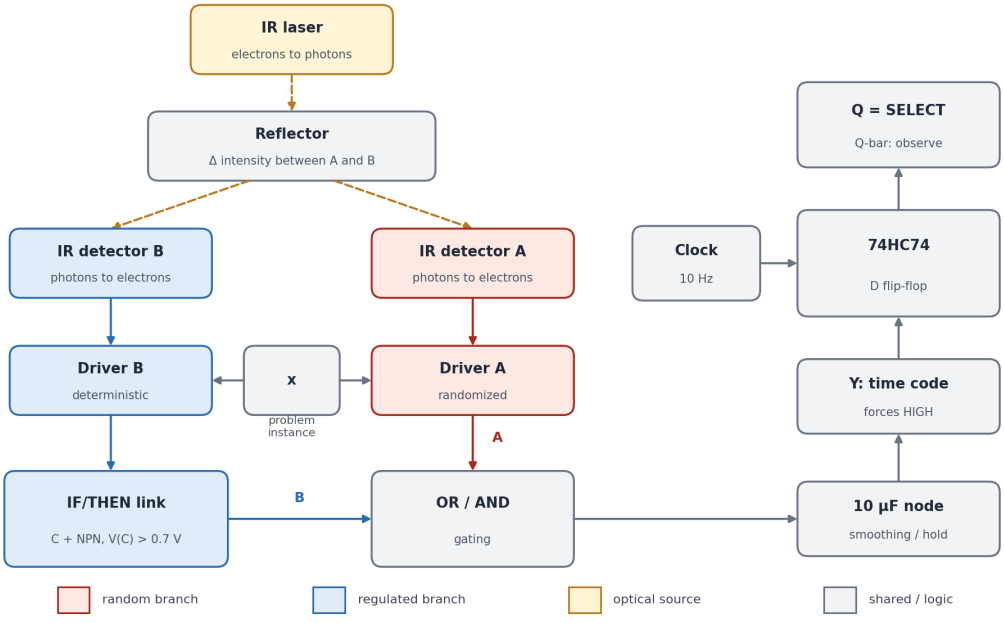
add windows. If each window has a false-fire probability ε, the chance of at least one false accept in k windows is $1 - (1 - \varepsilon)^k$, roughly $k\varepsilon$ for small ε.

6. Walking through the re-labelled circuit

1. **Emit.** The laser, driven by electrons, emits photons. Their Fano factor is set by how regular the pump is.
2. **Split.** The reflector sends the beam to detectors A and B. Each arm is a thinned copy, and the intensity difference between them carries the signal.
3. **Detect.** Each detector turns photons into electrons and thins the stream again. Detector A feeds the random Driver A, detector B feeds the regulated Driver B.
4. **Integrate.** In the IF/THEN link, C accumulates charge from Driver B. When V(C) passes about 0.7 V the NPN passes B HIGH. Averaging over millions of electrons makes the crossing nearly deterministic.
5. **Gate.** The AND stage lets the result through only when A and B coincide. The OR stage lets either one through.
6. **Smooth and time.** The 10 µF node filters fast noise and the time code Y marks the allowed bins.
7. **Detect and latch.** On each clock edge the flip-flop records click or no click.
8. **Accumulate.** The histogram of clicks over many windows is the photon-number distribution, and its Fano factor reveals which electron statistics were in charge.

7. Integration: one chain from electrons to counted photons

Putting the circuit and the physics together gives a single chain. Pump electrons drive the laser, the photons are split and thinned by the reflector, and the detectors turn them back into electrons, thinned once more. The two electron streams then feed a random channel and a regulated channel in parallel. The regulated one passes through the charge-threshold link. Their outputs are gated, smoothed by the 10 µF node and sliced into time bins by the code Y, and the flip-flop turns the result into a click. Unlike the earlier drawing, no path leads from the output back to the pump, so the chain is feed-forward. Adding a wire from Q back to the laser drive would be the standard route to pump-noise suppression, but it is an optional extension and is not in this drawing.



Feed-forward only: no path leads from Q back to the drivers, the laser or the clock.

Integrated block diagram: red path = random branch, blue = regulated branch, yellow = optical source

Operating mode	Which branch dominates	Photon statistics out
A only	Random injection, no regulation	$F_{ph} = 1$ (Poissonian), or above 1 with thermal mixing.
B only, through the IF/THEN link	Regulated injection, averaged on C	$F_{ph} = 1 - \eta$ (sub-Poissonian), with the threshold crossing nearly deterministic.
A and B with AND gating	Result only when both conditions coincide	Intermediate: noise from A is limited by the regulated gate from B.
A and B with feedback from Q (extension, not drawn)	Output corrects the pump	F_{ph} pushed below 1, limited by η and loop bandwidth (set by the RC node).

8. Limits of the analogy

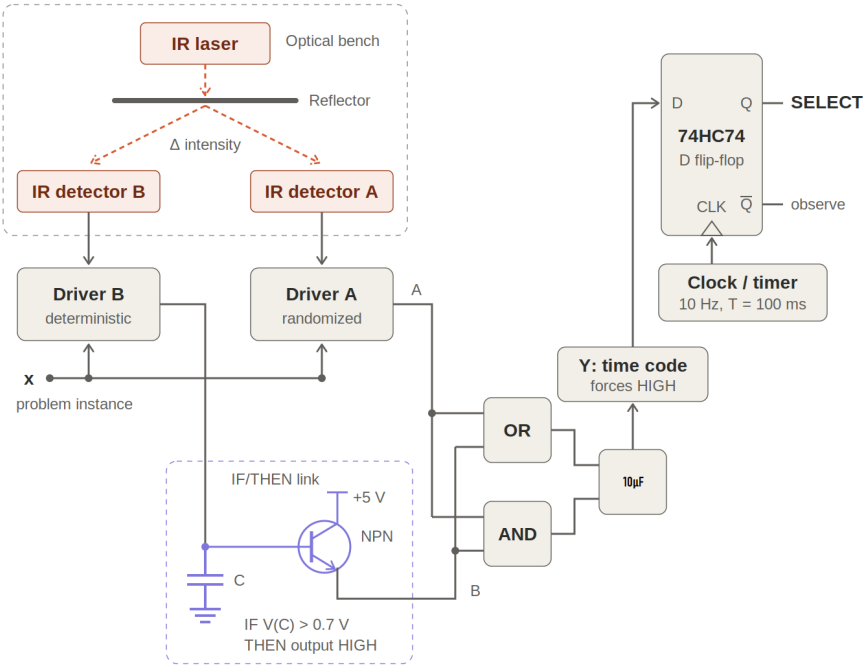
- The circuit is a *decision* device with a single bit out. Photon statistics are a property of many windows, so the mapping is conceptual, not a circuit you could build to emit light.
- The relation $F_{ph} = 1 - \eta(1 - F_e)$ assumes independent loss and one photon per electron. Real devices add correlations from recombination dynamics, cavity effects and detector dead time.
- The drawing's gate wiring (OR and AND outputs tied together) is ambiguous: as a plain OR it would allow false accepts. Only the reading in which both conditions must hold, which reduces to A AND B and keeps the one-sided guarantee, is used here.
- Both detectors see the same laser, so laser intensity noise is common to both branches. Balanced detection would subtract it, but here each detector feeds its own driver, so A and B are not fully independent.
- The 0.7 V rule in the IF/THEN link is the transistor's turn-on voltage, not a logic level (Part 1, section 4).
- Strongly sub-Poissonian emission in practice needs low temperature, high efficiency and low noise. Real LEDs and lasers usually sit near $F = 1$.

Summary: random electrons give random photons, regulated electrons give regular photons, and every loss or gate between the two pulls the distribution toward Poisson. The circuit's laser and detectors, random driver, deterministic driver with its charge threshold, gating, smoothing, timing and latch map onto exactly those steps.

PART 3 OF 3

The Protocol: Energy-Level Switching, Measurement and Closed Timelike Curves

How one trial of the optoelectronic RP-style circuit works as a physical protocol, and what a feedback loop would have to do with time loops



The circuit this protocol is built on

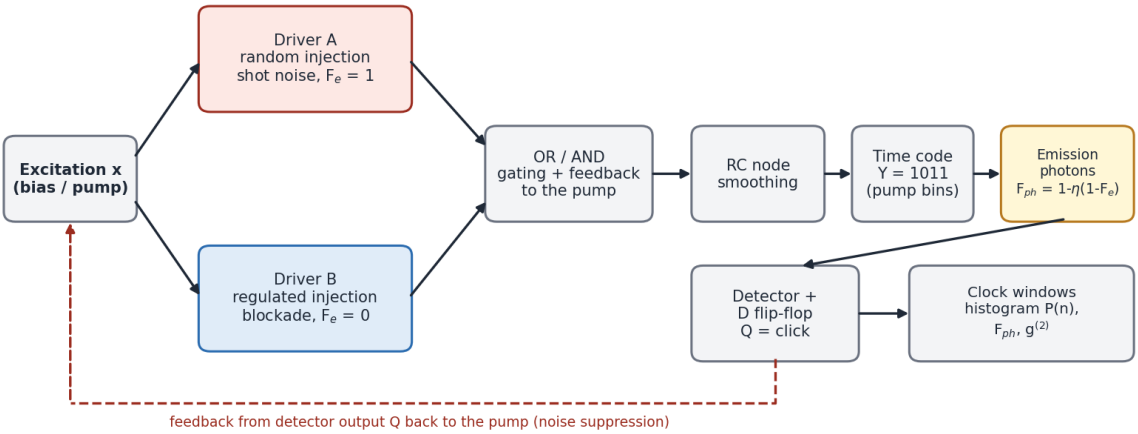
1. The protocol in six steps

Step	Circuit element	Physical event	Result
1. Prepare	Input x, laser drive, clock, time code Y	Choose the excitation, the reflector setting and the time bins in which the result may pass.	Defines when and how strongly electrons are driven.
2. Switch levels	IR laser, detectors A and B	In the laser an electron drops from E_2 to E_1 and a photon with $h\nu = E_2 - E_1$ leaves. In each detector an absorbed photon lifts an electron across the gap.	Photons emitted, random or regular, then turned back into electrons.
3. Threshold	Drivers A and B, IF/THEN link	A tests a witness on its noisy signal. B's current charges C until V(C) passes about 0.7 V and the NPN conducts.	Bits A and B.
4. Gate	OR / AND gating	The output is allowed only when the required conditions coincide.	Combined condition, read here as A AND B.
5. Measure	10 µF node, Y, 74HC74, CLK	The node smooths the result, Y marks the allowed bin, and on the clock edge the flip-flop latches it.	Q = SELECT = 1 (accept) or 0.
6. Repeat	Clock windows	Run k independent trials; accept if any trial gives Q = 1.	Worst-case error $(1/2)^k$.

2. Photons controlled by electron energy-level switching

A photon is emitted when an electron drops between two energy levels, and its energy is fixed by the gap. The same rule runs in reverse at a detector, where an absorbed photon lifts an electron across a gap. The optical bench therefore contains two level switches in series. What the circuit's drivers control or exploit is *when* the switches happen, and that sets the photon statistics.

- **IR laser, emission.** Stimulated emission above threshold gives coherent light with Poissonian counts (Fano factor F = 1). Thermal light gives F above 1.
- **Detectors A and B, absorption.** Each absorbed photon releases an electron with probability η, so the photocurrent inherits the photon statistics. Driver A uses the resulting noise as its random source.
- **Driver B, deterministic.** The IF/THEN link averages many electrons on C before the NPN conducts, so its crossing time barely fluctuates. On the emission side the deterministic counterpart is a resonant pulse with area π, which inverts a two-level system with probability close to 1, since the excited-state probability is $\sin^2(\theta/2)$. After emitting, the emitter is in its ground state and cannot emit again until it is re-excited, so photons tend to arrive one at a time. This is antibunching, with $g^{(2)}(0)$ approaching 0 and F < 1.
- **Clock and time code.** They decide which bins are pumped or counted, so they fix the counting window and the temporal pattern of the emission.



Left: the pump moves an electron from E_1 to E_2, and relaxation emits a photon. Right: the correlation function $g^{(2)}(\tau)$ for three kinds of source.

3. What measurement does

Photon-number outcomes follow the Born rule, $P(n) = |\langle n|\psi\rangle|^2$. Right after emission the emitter and the photon can be entangled, so detecting the photon also updates what you know about the emitter. The measurement chain adds its own effects:

- A detector with efficiency η thins the stream, so $F_{ph} = 1 - \eta(1 - F_e)$. Regularity from Driver B survives only as far as η allows.
- The click is an irreversible, thresholded event. This circuit has two such steps: the NPN turn-on at about 0.7 V and the flip-flop's sampling at the clock edge. A clocked flip-flop is a fair model of the second: it samples at the clock edge and holds the result.
- Noise and dark counts on the A side can fire Driver A when there is no witness. They are false accepts. The ideal RP circuit has none, so this is where the analogy breaks.

4. A feedback loop and closed timelike curves

The circuit as drawn has no feedback. Q goes to SELECT and Q-bar to “observe”. Nothing returns to x, the drivers, the laser or the clock, so the signal flow is strictly feed-forward. The earlier version of the figure carried a note about swapping A or B with Q, which did describe a loop. This version drops it. The link to closed timelike curves therefore exists only as an extension: it appears if you add a wire from Q (or Q-bar) back into the drivers or the laser drive.

A **closed timelike curve (CTC)** is a worldline that returns to its own past. CTCs appear as solutions of general relativity (for example Gödel’s rotating universe and some wormhole geometries). There is no evidence that they exist, and Hawking’s chronology protection conjecture argues that physics forbids them.

Deutsch’s model. In David Deutsch’s 1991 quantum model, a system going around a CTC must come back in the same state it left in. The state ρ of the looping system satisfies a fixed-point condition:

$$\rho_{CTC} = \mathrm{Tr}_{other}\left[U\left(\rho_{other} \otimes \rho_{CTC} \right) U^{\dagger} \right]$$

Grandfather paradox as a bit. If the loop must flip a bit (x = NOT x), no definite value is consistent. Deutsch’s rule picks the 50/50 mixture, which is the only self-consistent state. It echoes the 1/2 success threshold of the randomized driver, but this is an analogy and not a derivation.

An ordinary loop resolves it with delay. Wire Q-bar back to D on the 74HC74 and the same x = NOT x condition appears, but the flip-flop only updates at clock edges. Each edge flips the output, so Q toggles and runs at 5 Hz from the 10 Hz clock. The contradiction is resolved by the delay of one clock period, which a CTC does not allow.

Complexity. Aaronson and Watrous (2009) showed that with CTCs, both classical and quantum computers would solve exactly the problems in PSPACE, a class that contains RP.

Experiments. Ringbauer and colleagues (2014) simulated CTC behaviour with photons using post-selection. This reproduces the mathematics, not an actual loop through time.

> In this circuit there is no loop at all. If one were added it would be an ordinary electrical loop, which settles in ordinary time, not a time loop. It would resemble the CTC fixed-point condition only in that the output must be consistent with the input it helps produce.

5. Parameters of the system

Parameter	Role	Where it appears
$E_2 - E_1$	Photon energy and wavelength (IR here)	Laser emission and detector absorption
τ (emitter lifetime)	Emission rate; recovery time between photons	$g^{(2)}(\tau)$ dip width
Pulse area θ	How deterministic the excitation is (π gives inversion)	Emission side of Driver B
F_e, η_{emit}, T, η_d	Electron noise, laser efficiency, share of light reaching the detector, detector efficiency, giving F_{ph} and F_{det}	Part 2, section 4
Reflector split (Δ intensity)	Share of the light reaching A and B	Optical bench
C and the 0.7 V threshold	Charge N = CV/e needed before the NPN conducts, and the time to get there	IF/THEN link
Noise and dark-count rate	Sets the false-accept rate	Detector A and flip-flop
10 µF node	Smoothing time constant, together with its drive resistance	Gate outputs to Y

Parameter	Role	Where it appears
Clock period T	Counting window; 100 ms here, ns or shorter in real devices	CLK
Time code Y	Which bins let the result through	Time code stage
k (trials)	Number of repetitions; error $(1/2)^k$	Repeat step
Loop consistency	Fixed-point condition on a fed-back state (only if a feedback wire is added)	CTC reading of any feedback

6. What is established and what is speculative

Status	Content
Established physics	Level-switching emission and absorption; Poissonian, thermal and antibunched statistics; the Born rule; the thinning relation for efficiency; capacitor charging to a threshold; one-sided error and $(1/2)^k$ repetition.
Model-dependent	Deutsch's fixed-point rule for CTCs. Other models (post-selected CTCs, for example) give different predictions, and Deutsch's model has been criticised for its nonlinear features.
Speculative	Whether CTCs exist at all, and any link between a feedback wire added to this circuit and a real time loop.

7. References to check

- D. Deutsch, "Quantum mechanics near closed timelike lines", Physical Review D, 1991.
- S. Aaronson and J. Watrous, "Closed timelike curves make quantum and classical computing equivalent", Proceedings of the Royal Society A, 2009.
- M. Ringbauer et al., "Experimental simulation of closed timelike curves", Nature Communications, 2014.
- S. Hawking, "Chronology protection conjecture", Physical Review D, 1992.

These citations are from memory and have not been checked against the publications.